

SUNDIALS

Suite of Nonlinear and Differential
/Algebraic Equation Solvers



Adaptive time integrators for ODEs and DAEs and efficient nonlinear solvers
Used in a variety of applications. Freely available. Encapsulated solvers & parallelism.

- **ODE and DAE time integrators:**

- *CVODE*: adaptive order and step BDF (stiff) & Adams (non-stiff) methods for ODEs
- *ARKODE*: adaptive step implicit, explicit, ImEx, and multirate Runge-Kutta methods for ODEs
- *IDA*: adaptive order and step BDF methods for DAEs
- *CVODES* and *IDAS*: provide forward and adjoint sensitivity analysis capabilities

- **Nonlinear Solvers:** *KINSOL* – Newton-Krylov; accelerated Picard and fixed point

- **Modular Design:** Easily incorporated into existing codes; Users can supply their own data structures and solvers or use SUNDIALS provided modules

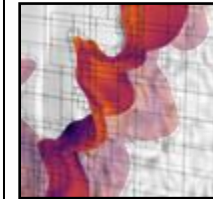
- **Supports laptops to GPU accelerated supercomputers:**

- Vectors: CUDA, HIP, SYCL, Kokkos, RAJA, OpenMP, MPI
- Linear solvers: MAGMA, Ginkgo, SuperLU_DIST, matrix-free Krylov methods

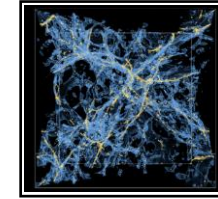
- **Open Source:** BSD License; Download from LLNL site, GitHub, or Spack

- Supported by extensive documentation; user email list with an active community
- Available through MFEM, AMReX, deal.II, and PETSc

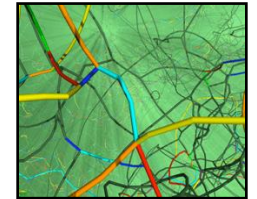
***SUNDIALS is used worldwide in applications
throughout research and industry***



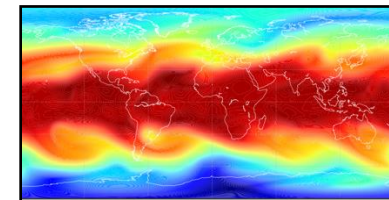
Combustion
(Pele)



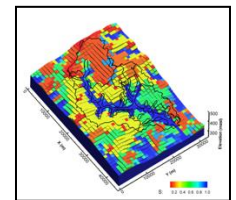
Cosmology
(Nyx)



Dislocation dynamics
(ParaDiS)



Atmospheric Dynamics
(Tempest)



Subsurface flow
(ParFlow)



<http://www.llnl.gov/casc/sundials>